Data Mining for Business Analytics
MSBA 511

Naïve Bayes Classifier

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Naïve Bayes Classification

The Naïve Bayes algorithm is a powerful probabilistic classifier based on Bayes' Theorem. Considered **"naïve"** since it assumes all features independent of each other, given the class label.

Simple Idea: Classify a record like records with the same values as the predictors

Some key points:

- Another **data driven**, not model driven algorithm.
- **Non-parametric** so it makes no assumptions about the data distribution.
- Requires **categorical** variables.
  - Numerical variables must be binned and converted to categorical.
- Works well with text data commonly used in spam detection, sentiment analysis, and document classification.

Exact Bayes Classifier

Relies on finding other records that share same predictor values as record-to-be-classified.

Want to find "probability of belonging to class *C*, given specified values of predictors."

Even with large data sets, may be hard to find other records that **exactly match** your record, in terms of predictor values.

Solution – Naïve Bayes

- Assume independence of predictor variables (within each class).
- Use multiplication rule
- Find same probability that record belongs to class C, given predictor values, <u>without</u> limiting calculation to records that share all those same values.

Naïve Bayes Calculations

1. Take a record and note its predictor values.
2. Find the probabilities those predictor values occur across all records in C1.
3. Multiply them together, then by proportion of records belonging to C1.
4. Same for C2, C3, etc.
5. Prob. of belonging to C1 is value from step (3) divide by sum of all such values C1 … Cn.
6. Establish & adjust a “cutoff” prob. for class of interest.

Example: Financial Fraud

Target variable: Audit finds fraud, no fraud

Predictors:

- Prior Legal Trouble (Yes/No)
- Company Size (Small/Large)

Company	Prior Legal Trouble	Company Size	Audit Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	Small	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Exact Bayes Calculations

Classify a small firm with prior legal trouble.

$$P(Fraud \mid Legal\ Trouble = Yes, Size = Small)$$

$$= \frac{\mathbf{1}}{\mathbf{2}} = \mathbf{0.50}$$

Calculation is limited to the two firms matching those characteristics.

Company	Prior Legal Trouble	Company Size	Audit Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	Small	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Naive Bayes Calculations

Classify a small firm with prior legal trouble.

$$P(Fraud \mid Legal\ Trouble = Yes, Size = Small)$$

Step 2: Find the probabilities for Legal Trouble = Yes, Size = Small given Fraud.

But first let's calculate the probability of Fraud.

$$\boldsymbol{P(Fraud) = \frac{4}{10} = .40}$$

Company	Prior Legal Trouble	Company Size	Audit Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	Small	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Naive Bayes Calculations

Classify a small firm with prior legal trouble.

$$P(Fraud \mid Legal\ Trouble = Yes, Size = Small)$$

Step 2: Find the probabilities for Legal Trouble = Yes, Size = Small given Fraud.

$$\boldsymbol{P(Fraud) = \frac{4}{10} = 0.40}$$

$$\boldsymbol{P(Legal \mid Fraud) = \frac{3}{4} = 0.75}$$

$$\boldsymbol{P(Small \mid Fraud) = \frac{1}{4} = 0.25}$$

Step 3: Multiply them together, then by proportion of records belonging to Fraud.

$$\boldsymbol{= 0.75 * 0.25 * 0.40 = 0.075}$$

Company	Prior Legal Trouble	Company Size	Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	Small	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Naive Bayes Calculations

Classify a small firm with prior legal trouble.

$$P(Fraud \mid Legal\ Trouble = Yes, Size = Small)$$

Step 4: Repeat for Truthful.

$$P(Truthful) = 1 - 0.40 = 0.60$$

$$P(Legal \mid Truthful) = \frac{1}{6} = 0.167$$

$$P(Small \mid Truthful) = \frac{4}{6} = 0.667$$

Repeat Step 3: Multiply them together, then by proportion of records belonging to Truthful.

$$= 0.167 * 0.667 * 0.60 = 0.067$$

Company	Prior Legal Trouble	Company Size	Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	Small	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Naive Bayes Calculations

Classify a small firm with prior legal trouble.

$$P(Fraud \mid Legal\ Trouble = Yes, Size = Small)$$

Step 5: Prob. of belonging to Fraud is value from Step 3 divided by sum of Fraud and Truthful.

$$= \frac{\mathbf{0.075}}{(\mathbf{0.075} + \mathbf{0.067})} = \mathbf{0.53}$$

Company	Prior Legal Trouble	Company Size	Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	Small	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Naive Bayes Continued

- Note that probability **estimate** does not differ greatly from **exact** but may not always be the case.

- All records are used in calculations, not just those matching predictor values.

- This makes calculations practical in most circumstances.

- Relies on assumption of independence between predictor variables within each class
  - Not strictly justified since variables are often correlated with one another.
  - Often "good enough" where ranking of probabilities is more important than unbiased estimate of actual probabilities.

One Important Consideration

What would happen with our probabilities and classification for a Small company if all companies in the training data were Large?

$$P(Small \mid Fraud) = \frac{0}{4} = 0.00$$

The absence of this predictor actively "outvotes" any other information in the record when making a classification.

Company	Prior Legal Trouble	Company Size	Outcome
1	Yes	Small	Truthful
2	No	Small	Truthful
3	No	Large	Truthful
4	No	Large	Truthful
5	No	Small	Truthful
6	No	Small	Truthful
7	Yes	**Large**	Fraud
8	Yes	Large	Fraud
9	No	Large	Fraud
10	Yes	Large	Fraud

Smoothing Techniques

To solve for this, a method called Laplace Smoothing (additive smoothing) adds a small positive value (typically 1) to the count of each category in the feature, ensuring no probability is ever exactly zero.

$$P(x_i \mid y) = \frac{count(x_i, y) + a}{N + a * K}$$

Alpha is the smoothing parameter

The sklearn implementation of MultinomialNB utilizes the default additive smoothing parameter with an alpha = 1.0.

Let's Practice: Personal Loan Example

In this example, we will work with only 2 binary variables and the target variable. The goal is to help a bank improve its conversion rate for personal loan acceptance in its next marketing campaign. We are working with training data for **3000** customers and the predictor variables are whether the customer currently has a Credit Card and Online Banking.

Here is a summary of the data:

Credit Card (CC)	Online (OL)	Personal Loan (Loan)	
		0	1
0	0	792	73
	1	1117	126
1	0	327	39
	1	477	49

Let's Practice: Personal Loan Example

Calculate the exact probability.

Credit Card (CC)	Online (OL)	Personal Loan (Loan)	
		0	1
0	0	792	73
	1	1117	126
1	0	327	39
	1	477	49

$$P(Loan = 1 \mid CC = 1, OL = 1)$$

$$= \frac{\mathbf{49}}{\mathbf{(49 + 477)}}$$

$$= \mathbf{0.093}$$

Let's Practice: Personal Loan Example

Calculate the Naïve Bayes probability.

Credit Card (CC)	Personal Loan (Loan)	
	0	1
0	1909	$\boxed{199}$
1	804	$\boxed{88}$

Online (OL)	Personal Loan (Loan)	
	0	1
0	1119	$\boxed{112}$
1	1594	$\boxed{175}$

$$P(Loan = 1) = \frac{287}{3000} = 0.095667$$

$$P(CC \mid Loan = 1) = \frac{88}{(199+88)} = 0.30662$$

$$P(OL \mid Loan = 1) = \frac{175}{(112+175)} = 0.609756$$

$$= 0.0957 * 0.307 * 0.61$$

$$= 0.0179$$

Let's Practice: Personal Loan Example

Calculate the Naïve Bayes probability.

Credit Card (CC)	Personal Loan (Loan)	
	0	1
0	$\boxed{1909}$	199
1	$\boxed{804}$	88

Online (OL)	Personal Loan (Loan)	
	0	1
0	$\boxed{1119}$	112
1	$\boxed{1594}$	175

$$P(Loan = 0) = \frac{2713}{3000} = 0.904333$$

$$P(CC \mid Loan = 0) = \frac{804}{(1909+804)} = 0.296351$$

$$P(OL \mid Loan = 0) = \frac{1594}{(1119+1594)} = 0.587541$$

$$= 0.904 * 0.296 * 0.588$$

$$= 0.157$$

$$P(Loan = 1 \mid CC = 1, OL = 1) = \frac{0.0179}{(0.0179+0.157)} = 0.102$$

Advantages and Shortcomings

Advantages

- Handles purely categorical data well.
- Works well with very large data sets.
- Simple and computationally efficient.

Shortcomings

- Requires a large training dataset.
- Requires smoothing techniques to handle when important rare events are not present in the training data.